

$$f(D)y = X$$

$$P.I = y_p = \frac{1}{f(D)} X$$

$$\text{Type ① } X = e^{ax}$$

$$y_p = \frac{1}{f(D)} e^{ax}$$

$$= \frac{1}{f(a)} e^{ax}$$

If $f(a) = 0$ then

$$y_p = x \frac{1}{f'(D)} e^{ax}$$

$$= x \frac{1}{f'(a)} e^{ax} \quad \text{--- } f'(a) \neq 0$$

If $f'(a) = 0$ then

$$y_p = x^2 \frac{1}{f''(D)} e^{ax}$$

$$= x^2 \frac{1}{f''(a)} e^{ax} \quad \text{--- } f''(a) \neq 0$$

$$\frac{e^{ax} + e^{-ax}}{2}$$

$$\frac{e^{ax} - e^{-ax}}{2}$$

$$X = \cosh ax / \sinh ax$$

$$X = C = C e^{0x}$$

$$X = a^x = e^{\log a^x} = e^{x(\log a)}$$

$$f(a) \neq 0$$

Type ② $X = \sin(ax+b) / \cos(ax+b)$

$$y_p = \frac{1}{f(D)} \sin(ax+b)$$

$$= \frac{1}{\phi(D^2)} \sin(ax+b) \quad \left(\text{substitute every } D^2 \text{ by } -a^2 \right)$$

$$= \frac{1}{\phi(-a^2)} \sin(ax+b) \quad \text{--- } \phi(-a^2) \neq 0$$

∴ $\phi(-a^2) = 0$

$$y_p = x \frac{1}{\phi(D^2)} \sin(ax+b)$$

$$= x \frac{1}{\phi'(-a^2)} \sin(ax+b) \quad \text{--- } \phi'(-a^2) \neq 0$$

Type 3 $X = x^m$ ^{m is +ve int}

$$y_p = \frac{1}{f(D)} x^m$$

$$\frac{D^2 + 1 + 2D}{1 + 2D + D^2}$$

write $f(D)$ in increasing power of D

$$= \frac{1}{[1 + \phi(D)]} x^m$$

in $\phi(D)$ there should not be any const

$$\frac{1}{1 + \phi(D)} = 1 - \phi(D) + [\phi(D)]^2 - \dots$$

$$\frac{1}{1 - \phi(D)} = 1 + \phi(D) + [\phi(D)]^2 + \dots$$

$$\frac{1}{1 + (1 + D + D^2)^2}$$

$$\frac{1}{2 \left(1 + \frac{D}{2} + \frac{D^2}{2} \right)}$$

$$\frac{1}{2 + D + D^2} = \frac{1}{1 + (1 + D + D^2)}$$

$$\frac{1}{2 \left(1 + \frac{D}{2} + \frac{D^2}{2} \right)}$$

$$\frac{1}{2} \left[1 + \frac{D}{2} + \frac{D^2}{2} + \left(\frac{D}{2} + \frac{D^2}{2} \right)^2 + \dots \right] (x^3)$$

$$1 - (1 + D + D^2) + (1 + D + D^2)^2 - \dots$$

Type 4

$$X = e^{ax} v$$

where v is funⁿ of x .

$$y_p = \frac{1}{f(D)} e^{ax} v$$

$$= e^{ax} \frac{1}{f(D+a)} v$$

❖ Solve $(D^3 - 1)y = (e^x + 1)^2$

$$(D^3 - 1)y = 0$$

$$A - eq^n \quad D^3 - 1 = 0$$

$$D = 1, -\frac{1}{2} \pm i\frac{\sqrt{3}}{2}$$

$$a = -\frac{1}{2}, b = \frac{\sqrt{3}}{2}$$

$$y_c = c_1 e^x + e^{-\frac{1}{2}x} \left(c_2 \cos \frac{\sqrt{3}}{2}x + c_3 \sin \frac{\sqrt{3}}{2}x \right)$$

$$y_p = \frac{1}{D^3 - 1} (e^x + 1)^2$$

$$= \frac{1}{D^3 - 1} (e^{2x} + 2e^x + 1)$$

$$= \frac{1}{D^3 - 1} e^{2x} + 2 \frac{1}{D^3 - 1} e^x + \frac{1}{D^3 - 1} e^{0x}$$

$$= \frac{1}{2^3 - 1} e^{2x} + 2x \frac{1}{3D^2} e^x + \frac{1}{-1}$$

$$= \frac{1}{7} e^{2x} + \frac{2x}{3} e^x - 1$$

$$y = y_c + y_p$$

$$= c_1 e^x + e^{-\frac{x}{2}} \left(c_2 \cos \frac{\sqrt{3}}{2}x + c_3 \sin \frac{\sqrt{3}}{2}x \right) + \frac{e^{2x}}{7} + \frac{2xe^x}{3} - 1$$

Solve $6 \frac{d^2 y}{dx^2} + 17 \frac{dy}{dx} + 12y = e^{-\frac{3}{2}x} + 2^x$

$$6D^2 y + 17Dy + 12y = e^{-\frac{3}{2}x} + 2^x$$

$$(6D^2 + 17D + 12)y = e^{-\frac{3}{2}x} + 2^x$$

A. eqⁿ $6D^2 + 17D + 12 = 0$

$$D = -\frac{4}{3}, -\frac{3}{2}$$

$$y_c = C_1 e^{-\frac{4}{3}x} + C_2 e^{-\frac{3}{2}x}$$

$$y_p = \frac{1}{6D^2 + 17D + 12} (e^{-\frac{3}{2}x} + 2^x)$$

$$= \frac{1}{6D^2 + 17D + 12} e^{-\frac{3}{2}x} + \frac{1}{6D^2 + 17D + 12} e^{(\log 2)x}$$

$6D^2 + 17D + 12 = 0$
for $D = -\frac{3}{2}$

$$= x \frac{1}{12D + 17} e^{-\frac{3}{2}x} + \frac{1}{6(\log 2)^2 + 17\log 2 + 12} e^{(\log 2)x}$$

$$= x \frac{1}{6(12(-\frac{3}{2}) + 17)} e^{-\frac{3}{2}x} + \frac{2^x}{6(\log 2)^2 + 17\log 2 + 12}$$

$$= -x e^{-\frac{3}{2}x} + \frac{2^x}{6(\log 2)^2 + 17\log 2 + 12}$$

$$y = y_c + y_p$$

Solve $(D^2 + 4D + 4)y = \cosh 2x$

$$= \frac{e^{2x}}{2} + \frac{e^{-2x}}{2}$$

$$y_p = \frac{1}{D^2 + 4D + 4} \frac{e^{2x}}{2}$$

$$= \frac{1}{2} \frac{1}{D^2 + 4D + 4} e^{2x}$$

$$= \frac{x}{2} \frac{1}{2D + 4} e^{2x}$$

$$\rightarrow \frac{x^2}{2} \frac{1}{2} e^{2x}$$

$$(\operatorname{cosec} x) \frac{d^2 y}{dx^2} + (\operatorname{cosec} x) y = \sin 2x$$

$$D^2 y + y = \frac{\sin 2x}{\operatorname{cosec} x}$$

$$(D^2 + 1) y = \sin 2x \sin x$$

$$A = e^{ax} \quad D^2 + 1 = 0 \quad D = \pm i \quad a = 0 \quad b = 1$$

$$y_e = C_1 \cos x + C_2 \sin x$$

$$y_p = \frac{1}{D^2 + 1} \sin 2x \sin x$$

$$= \frac{1}{D^2 + 1} \left(\frac{\cos x - \cos 3x}{2} \right)$$

$$= \frac{1}{2} \frac{1}{D^2 + 1} \cos x - \frac{1}{2} \frac{1}{D^2 + 1} \cos 3x$$

$$2 \sin A \cos B = \sin(A+B) + \sin(A-B)$$

$$2 \cos A \sin B = \sin(A+B) - \sin(A-B)$$

$$2 \cos A \cos B = \cos(A+B) + \cos(A-B)$$

$$2 \sin A \sin B = \cos(A-B) - \cos(A+B)$$

$$D^2 + 1 = 0$$

$$\text{for } D^2 = -1$$

$$= \frac{x}{2} \frac{1}{2D} \cos x - \frac{1}{2} \frac{1}{(-9+1)} \cos 3x$$

$$= \frac{x}{4} \int \cos x \, dx + \frac{1}{16} \cos 3x$$

$$= \frac{x}{4} \sin x + \frac{1}{16} \cos 3x$$

$$y = y_e + y_p$$

HW

❖ Solve $(D^2 + 4)y = \cos(2x + 3)$

$$D^2 = -4$$

$$y_p = \frac{1}{D^2 + 4} \cos(2x + 3)$$

$$= \frac{1}{2D} \cos(2x + 3)$$

❖ Solve $(D-1)^2(D^2+1)y = e^x + \sin^2 \frac{x}{2}$

$$(D-1)^2(D^2+1) = 0$$

$$D = 1, 1, \pm i \quad a=0, \quad b=1$$

$$y_c = c_1 e^x + c_2 x e^x + c_3 \cos x + c_4 \sin x$$

$$y_p = \frac{1}{(D-1)^2(D^2+1)} \left(e^x + \sin^2 \frac{x}{2} \right)$$

$$= \frac{1}{(D-1)^2(D^2+1)} e^x + \frac{1}{(D-1)^2(D^2+1)} \sin^2 \frac{x}{2}$$

$$\frac{1}{(D-1)^2(D^2+1)} e^x = \frac{1}{(D-1)^2(2)} e^x$$

$$= \frac{x^2}{2} \frac{1}{2(D-1)} e^x = \frac{x^2}{2} \frac{1}{2} e^x = \frac{x^2 e^x}{4}$$

$$\frac{1}{(D-1)^2(D^2+1)} \sin^2 \left(\frac{x}{2} \right) = \frac{1}{(D-1)^2(D^2+1)} \left(\frac{1 - \cos x}{2} \right)$$

$$= \frac{1}{(D-1)^2(D^2+1)} \left(\frac{e^{0x} - \cos x}{2} \right)$$

$$\left. \begin{array}{l} \sin 1 + \cos 2\theta = 2 \cos^2 \theta \\ 1 - \cos 2\theta = 2 \sin^2 \theta \end{array} \right\} = \frac{1}{(D-1)^2(D^2+1)} \frac{e^{0x}}{2} - \frac{1}{2} \frac{1}{(D-1)^2(D^2+1)} \cos x$$

$$= \frac{1}{1} \frac{e^{0x}}{2} - \frac{1}{2} \frac{1}{(D^2+1)(D^2-2D+1)} \cos x$$

$$= \frac{1}{2} - \frac{1}{2} \frac{1}{(D^2+1)(-1-2D+1)} \cos x$$

$$= \frac{1}{2} + \frac{1}{2} \frac{1}{(D^2+1)D} \cos x$$

$$= \frac{1}{2} + \frac{1}{2} \frac{1}{D^3+D} \cos x$$

$$= \frac{1}{2} + \frac{x}{2} \left(\frac{1}{3D^2+1} \right) \cos x$$

$$= \frac{1}{2} + \frac{x}{4} \frac{1}{(-3)+1} \cos x$$

$$= \frac{1}{2} - \frac{x}{8} \cos x$$

$$\frac{1}{D^2+1} \int \cos x \, dx$$

$$\frac{1}{D^2+1} \sin x$$

$$x \frac{1}{2D} \sin x$$

$$-\frac{x}{2} \cos x$$

$$y = y_c + y_p$$

Solve $(D^3 + 1)y = \cos(2x - 1)$

HW

• Solve $(D^3 + D^2 + D + 1)y = \sin^2 x$ $= \left(\frac{1 - \cos 2x}{2} \right)$

$$\left(\frac{e^{0x} - \cos 2x}{2} \right)$$

Solve $(D^3 + 2D^2 + D)y = x^2 + x$

A. eqⁿ $D^3 + 2D^2 + D = 0$
 $D(D^2 + 2D + 1) = 0$
 $D(D+1)^2 = 0$
 $D = 0, -1, -1$

$$y_c = c_1 + c_2 e^{-x} + c_3 x e^{-x}$$

$$y_p = \frac{1}{D^3 + 2D^2 + D} (x^2 + x)$$

$$= \frac{1}{D + 2D^2 + D^3} (x^2 + x)$$

$$= \frac{1}{D} \frac{1}{(1 + 2D + D^2)} (x^2 + x)$$

$$= \frac{1}{D} \left[1 - (2D + D^2) + (2D + D^2)^2 - \dots \right] (x^2 + x)$$

$$= \frac{1}{D} \left[1 - 2D - D^2 + 4D^2 \right] (x^2 + x)$$

$$= \frac{1}{D} \left[1 - 2D + 3D^2 \right] (x^2 + x)$$

$$= \frac{1}{D} \left[x^2 + x - 2D(x^2 + x) + 3D^2(x^2 + x) \right]$$

$$= \frac{1}{D} \left[x^2 + x - 2(2x + 1) + 3(2) \right]$$

$$= \frac{1}{D} \left[x^2 - 3x + 4 \right]$$

$$= \int (x^2 - 3x + 4) dx = \frac{x^3}{3} - \frac{3x^2}{2} + 4x$$

$$y = y_c + y_p =$$

$$(x^2 + x + 2)$$

$$x^2 + x$$

$$2x + 1$$

$$2$$

$$3$$

$$0$$

Solve $(D^2 - 4D + 4)y = 8(x^2 + \sin 2x + e^{2x})$

$$(D^2 - 4D + 4)y = 0$$

A-49ⁿ $D^2 - 4D + 4 = 0$
 $D = 2, 2$ $y_c = C_1 e^{2x} + C_2 x e^{2x}$

$$y_p = \frac{1}{D^2 - 4D + 4} 8(x^2 + \sin 2x + e^{2x})$$

$$= \frac{1}{D^2 - 4D + 4} (8x^2) + \frac{1}{D^2 - 4D + 4} 8\sin 2x + \frac{1}{D^2 - 4D + 4} 8e^{2x}$$

$$\frac{1}{D^2 - 4D + 4} (8x^2) = 8 \frac{1}{4 - 4D + D^2} (x^2)$$

$$= \frac{8}{4} \left(\frac{1}{1 - D + \frac{D^2}{4}} \right) (x^2)$$

$$= 2 \frac{1}{1 - (D - \frac{D^2}{4})} x^2$$

$$= 2 \left[1 + D - \frac{D^2}{4} + (D - \frac{D^2}{4})^2 + \dots \right] x^2$$

$$= 2 \left[1 + D - \frac{D^2}{4} + D^2 \right] x^2$$

$$= 2 \left[1 + D + \frac{3D^2}{4} \right] x^2$$

$$= 2 \left[x^2 + 2x + \frac{3}{4}(2) \right]$$

$$= 2x^2 + 4x + 3$$

$\left| \begin{array}{l} x^2 \\ 2x \\ 2 \end{array} \right.$

$$\frac{1}{D^2 - 4D + 4} 8\sin 2x = 8 \frac{1}{D^2 - 4D + 4} \sin 2x$$

$$= 8 \frac{1}{-4 - 4D + 4} \sin 2x$$

$$= -2 \int \sin 2x \, dx = \frac{2 \cos 2x}{2}$$

$$\frac{1}{D^2 - 4D + 4} 8e^{2x} = 8 \frac{1}{D^2 - 4D + 4} e^{2x}$$

$$= 8x \frac{1}{2D - 4} e^{2x}$$

$$= \frac{4}{8} x^2 \frac{1}{2} e^{2x}$$

$$y_p = 2x^2 + 4x + 3 + \cos 2x + 4x^2 e^{2x}$$

$$y = y_c + y_p$$

❖ HW Solve $(D^3 - D)y = 2e^x + 2x + 1 - 4\cos x$

HW

$$\frac{1}{D^3 - D} (2e^x)$$

$$\frac{1}{D^3 - D} (2x + 1)$$

$$\frac{1}{D^3 - D} 4\cos x$$

❖ Solve $\frac{d^2y}{dx^2} + \frac{dy}{dx} - 6y = e^{2x} \sin 3x$

$$\frac{(x^2 + 2x + 4)}{\quad}$$

❖ HW Solve $(D^3 - D)y = 2e^x + 2x + 1 - 4\cos x$

$$2e^x$$

$$(2x + 1)$$

$$4\cos x$$

$$y_p = \frac{1}{f(D)} e^{ax}$$

$$= e^{ax} \frac{1}{f(D+a)}$$

❖ Solve $(D^2 + 4D + 4)y = \frac{e^{-2x}}{x^5}$

$$D^2 + 4D + 4 = 0 \quad D = -2, -2$$

$$y_c = C_1 e^{-2x} + C_2 x e^{-2x}$$

$$y_p = \frac{1}{D^2 + 4D + 4} \left(\frac{e^{-2x}}{x^5} \right)$$

$$= e^{-2x} \frac{1}{(D-2)^2 + 4(D-2) + 4} x^{-5}$$

$$= e^{-2x} \frac{1}{D^2 - 4D + 4 + 4D - 8 + 4} x^{-5}$$

$$= e^{-2x} \frac{1}{D} \int x^{-5} dx = e^{-2x} \frac{1}{D} \left(\frac{x^{-4}}{-4} \right)$$

$$= \frac{e^{-2x}}{-4} \frac{x^{-3}}{(-3)} = \frac{x^3 e^{-2x}}{12}$$

$$y = y_c + y_p$$

Solve $(D^2 + 2)y = e^x \cos x + x^2 e^{3x}$

$$D^2 + 2 = 0 \Rightarrow D = \pm i\sqrt{2} \quad \begin{matrix} a = 0 \\ b = \sqrt{2} \end{matrix}$$

$$y_c = C_1 \cos \sqrt{2}x + C_2 \sin \sqrt{2}x$$

$$y_p = \frac{1}{D^2 + 2} (e^x \cos x + x^2 e^{3x})$$

$$\frac{1}{D^2 + 2} e^x \cos x = e^x \frac{1}{(D+1)^2 + 2} \cos x$$

$$= e^x \frac{1}{D^2 + 2D + 3} \cos x$$

$$= e^x \frac{1}{-1 + 2D + 3} \cos x$$

$$= e^x \frac{1}{2 + 2D} \cos x$$

$$= \frac{e^x}{2} \frac{(D-1)}{(D+1)(D-1)} \cos x$$

$$= \frac{e^x}{2} \frac{(D-1)}{(D^2-1)} \cos x = \frac{e^x}{2} \frac{(D-1)}{-2} \cos x$$

$$= -\frac{e^x}{4} [-\sin x - \cos x] = \frac{e^x}{4} (\sin x + \cos x)$$

$$\frac{1}{D^2 + 2} x^2 e^{3x} = e^{3x} \frac{1}{(D+3)^2 + 2} x^2$$

$$= e^{3x} \frac{1}{D^2 + 6D + 11} x^2$$

$$= e^{3x} \frac{1}{11 + 6D + D^2} x^2$$

$$= \frac{e^{3x}}{11} \frac{1}{\left(1 + \frac{6}{11}D + \frac{D^2}{11}\right)} x^2$$

$$= \frac{e^{3x}}{11} \left[1 - \left(\frac{6}{11}D + \frac{D^2}{11}\right) + \left(\frac{6}{11}D + \frac{D^2}{11}\right)^2 - \right] x^2$$

$$= \frac{e^{3x}}{11} \left[1 - \frac{6}{11}D - \frac{D^2}{11} + \frac{36D^2}{121} \right] x^2$$

$$= \frac{e^{3x}}{11} \left[1 - \frac{6}{11}D + \frac{25D^2}{121} \right] x^2$$

$$= \frac{e^{3x}}{11} \left[x^2 - \frac{6}{11}(2x) + \frac{25}{121}(2) \right]$$

$$y_p = \frac{e^x}{4} (\sin x + \cos x) + \frac{e^{3x}}{11} \left[x^2 - \frac{12}{11}x + \frac{50}{121} \right]$$

$$y = y_c + y_p //$$